OCEAN STARTUP CHALLENGE · 2026

planetar.

Open, real-time maritime domain awareness — one live picture of *who is on the water*, including vessels trying not to be seen.

planetar.ca — live now | open source · 9 repos | Victoria, BC

Steven Ness, PhD · founder

Going dark works.

A vessel that wants to disappear just switches off its AIS transponder — and to the tools most ocean operators can afford, it does disappear. Illegal, unreported and unregulated fishing runs on exactly this gap.

AIS is voluntary

The one feed everyone watches can be turned off with a switch — no signal, no track.

Radar has gaps

Satellite radar sees dark vessels — but a pass every few hours, and small craft slip between revisits.

Cameras go blind

Electro-optical feeds fail at night and in weather — precisely when dark activity happens.

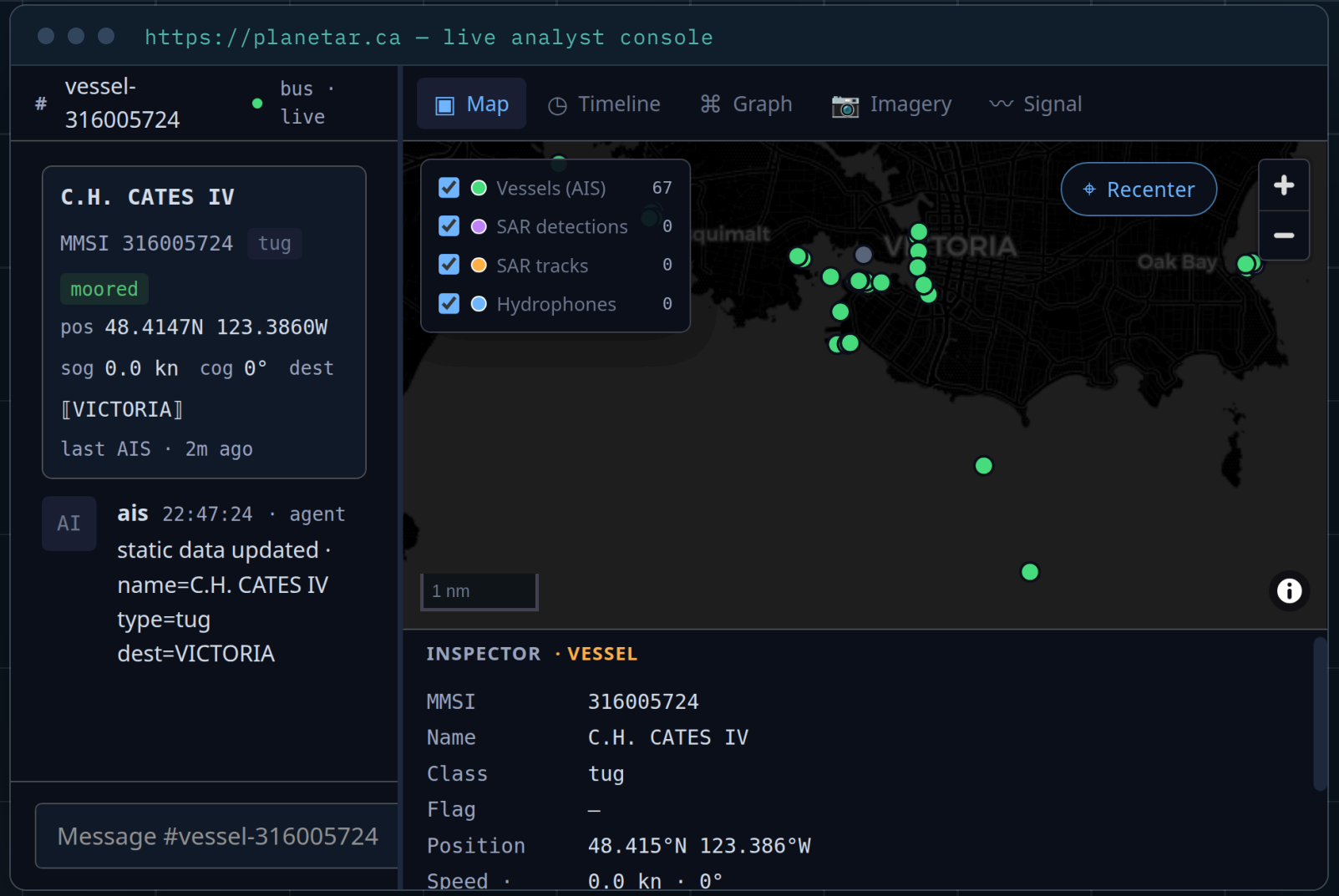
The platforms that fuse all of these exist — **closed, vendor-locked, and priced for navies and insurers**. Global Fishing Watch proved the demand for transparency, but it is global batch analysis, not a live local picture. Ports, fisheries officers and conservation groups are left watching one screen at a time.

Fuse every sensor. Lose no vessel.

Planetar fuses four live sensor streams over a low-latency message bus into one entity graph that re-identifies the same vessel across sensors — so switching off AIS doesn't make a vessel invisible, it makes it a correlated radar, camera and acoustic track.



100% open source · commodity hardware · public & low-cost data



▲ live right now: 67 vessels on the Salish Sea AIS layer · click any one for its full track and identity

24k+

lines of open-source code across nine public repositories (measured 2026-07-20)

4

sensor modalities running on real data — live AIS, Sentinel-1 radar, webcam EO, hydrophone acoustics

433Mpx

real Sentinel-1 radar scene the vessel-detection pipeline is validated on

planetar.ca

click it yourself — the demo is public and live

THE FOUNDER

A career spent listening to this ocean — then building the machine that watches it.

Ocean machine learning, peer-reviewed

PhD in computer science (University of Victoria) — published ML on Ocean Networks Canada hydrophone data, and co-author of ORCA-SLANG (Interspeech 2021), deep learning for killer-whale call identification.

The Orchive

Built the doctoral system behind a 20,000-hour, 30-year orca acoustic archive — archive-scale ML with a human-in-the-loop design.

Entity resolution at industry scale

Applicant-named inventor on US Patent 10,936,582 (integrated entity views; Salesforce-assigned) — the discipline behind Planetar's cross-sensor vessel re-identification.

The honest gap

Solo engineer-scientist who built the entire stack — what's missing is structured customer discovery and a first design partner. That is exactly what this cohort provides.

Priced out by closed platforms — served by open ones.

WHO NEEDS THIS

Fisheries enforcement — dark-vessel detection in closed and protected areas (the IUU gap)

Ports & harbour authorities — one live picture of approaches and anchorages

Conservation NGOs — marine-protected-area monitoring, whale-strike and pollution risk

Search & rescue — fused tracking of small and non-cooperative targets

HOW IT PAYS · OPEN SOURCE + SERVICES

1 · Pilots & integration

Paid design-partner deployments — the near-term revenue the $25K sprint feeds directly.

2 · Hosted service

Managed subscription for operators who don't run infrastructure.

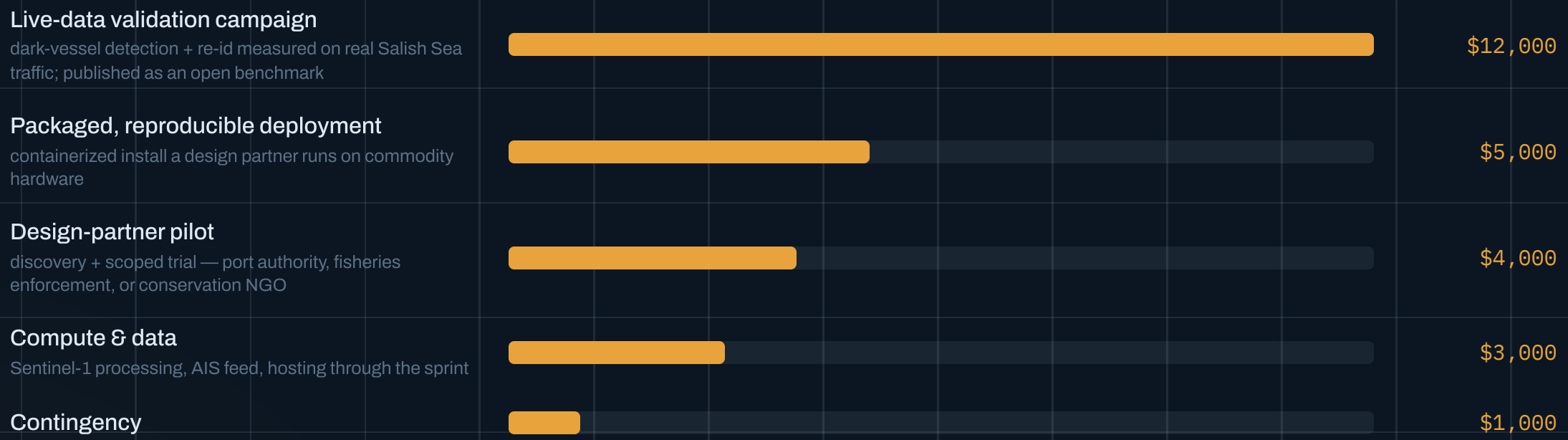
3 · Support agreements

GPLv3 platform — ongoing support and maintenance contracts as adoption grows; the open version stays free for public-interest users.

USE OF FUNDS

The $25K validation sprint

Three workstreams that turn a working system into a fundable, sellable one: a published benchmark, a package a partner can run, and a first design partner.



THE ASK

The ocean is watched by closed systems. Let's open it.

With the Ocean Startup Challenge: a published dark-vessel benchmark, a deployment any coastal operator can run, a first design partner — and a Canadian company built around open ocean infrastructure.

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